



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai 2026



ASSESSMENT TOOL -3

C04_AT3

Name : S Nikhileswar Reddy

Register No : 192525096

Course code : CSA1613

Course name: Data Warehousing and Data Mining

Faculty name: Dr.Kanimozhi KV

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Data Warehousing and Data Mining COURSE CODE: CSA1613

COURSE OUTCOME (CO4): Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Industry Problem Solving Task

Questions: 05

Total Marks: 50

Marks per Question: 7

Q1. Supermarket Purchases — Customer Segmentation for Targeted Marketing

Dataset:

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question: Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing.

Answer:

1. Understanding the Problem: The supermarket wants to group customers by shopping behaviour so that marketing campaigns (offers, coupons, loyalty rewards) can be tailored to each group rather than sent generically to everyone. The three behavioural attributes available — visit frequency, average spend, and items bought per visit — are enough to reveal distinct shopper types.
2. Application of Engineering Concepts: The data already shows two clear behavioural patterns. C1, C3 and C5 visit frequently (10–12 times/month), buy many items (15–18) but spend a moderate amount per

visit (₹1800–2100) — typical 'frequent, high-volume, budget-conscious' shoppers. C2 and C4 visit rarely (3–4 times/month) but spend heavily each time (₹5000–6000) on fewer items (6–8) — typical 'infrequent, high-value bulk buyers.' This is a natural clustering problem in 3-dimensional feature space (visits, spend, items).

3. Solution Design: Standardize the three features (z-score normalization) so that spend in rupees does not dominate visit-count purely due to scale, then apply K-means clustering with $k = 2$ (or use the Elbow Method / Silhouette Score to confirm k). K-means is appropriate here because the two groups are compact and well separated. Alternatively, hierarchical clustering could be used to visualize the segmentation via a dendrogram if the number of natural segments is uncertain.

4. Use of Tools: Implementation can be done in Python using pandas for data preparation, scikit-learn's StandardScaler and KMeans for clustering, and matplotlib/seaborn to visualize the resulting clusters (e.g., a scatter plot of spend vs. items bought, colour-coded by cluster).

5. Analysis & Business Impact: Cluster A (C1, C3, C5 — 'Frequent Value Shoppers') can be targeted with volume discounts, loyalty points, and bundle offers to encourage even more frequent visits. Cluster B (C2, C4 — 'High-Value Occasional Shoppers') can be targeted with premium product recommendations, personalized high-value coupons, and reminders to increase visit frequency. This segmentation lets the supermarket allocate marketing budget efficiently instead of using a one-size-fits-all campaign, directly improving conversion rates and customer lifetime value.

Q2. Credit Card Transactions — Fraud Detection

Dataset:

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question: Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions.

Answer:

1. Understanding the Problem: Fraudulent transactions typically differ sharply from a customer's normal spending pattern — unusually high amount, unfamiliar location, or an odd time of day. The goal is to automatically flag transactions that deviate from the customer's usual dense cluster of behaviour rather than manually reviewing every transaction.
2. Application of Engineering Concepts: In this sample, T1, T3 and T5 form a tight, dense cluster — small amounts (₹150–200), same location (Chennai), and daytime hours (10–12 hrs), representing normal behaviour. T2 and T4 are sparse, isolated points — very large amounts (₹5000, ₹7000), different cities (Delhi, Mumbai), and unusual early-morning hours (1–2 hrs). This is precisely the kind of pattern that density-based clustering is designed to detect.
3. Solution Design: Encode the categorical Location attribute (e.g., one-hot encoding) and normalize Amount and Time, then apply DBSCAN (Density-Based Spatial Clustering of Applications with Noise) with a suitable ϵ (neighbourhood radius) and minPts. DBSCAN will group T1, T3, T5 into one dense cluster (normal transactions) and mark T2 and T4 as noise points because they do not have enough neighbours within ϵ — exactly the outlier-flagging behaviour needed for fraud detection. K-means is deliberately avoided here because it forces every point (including genuine outliers) into a cluster and has no built-in concept of 'noise.'
4. Use of Tools: Python with scikit-learn's DBSCAN class, pandas for preprocessing, and a scatter plot (Amount vs. Time, coloured by cluster label, with noise points marked distinctly) to visually confirm which transactions were flagged.
5. Analysis & Business Impact: T2 and T4 would be automatically flagged as anomalies (label = -1 in DBSCAN) and routed to a fraud-review queue or trigger an SMS/OTP verification to the cardholder, while T1, T3, T5 pass through without friction. This reduces manual review workload, catches fraud in near real-time, and can be periodically re-run as new transactions arrive to adapt to evolving fraud patterns.

Q3. Telecom Usage — Churn-Prone User Identification

Dataset:

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question: Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies.

Answer:

1. Understanding the Problem: Telecom churn occurs when low-engagement, low-spending users switch providers. Identifying these users early — based on usage patterns rather than waiting for them to actually leave — allows the company to intervene with retention offers before churn happens.
2. Application of Engineering Concepts: U1, U3 and U5 show high call duration (300–350 min), high data usage (5–6 GB) and high recharge value (₹200–220) — clearly engaged, high-value users at low churn risk. U2 and U4 show low call duration (80–100 min), low data usage (1–2 GB) and low recharge (₹90–100) — a pattern strongly associated with disengagement and higher churn probability. These two behavioural groups are naturally separable in usage-feature space.
3. Solution Design: Standardize the three numeric features and apply K-means clustering with $k = 2$ to separate 'Engaged/Low-Risk' users from 'Low-Usage/High-Risk' users. Because churn risk in practice is rarely a hard binary and can shade gradually with usage decline, the K-means output can optionally be refined by computing each user's distance to the low-usage centroid as a continuous churn-risk score, or by using Gaussian Mixture Models (EM) to assign a probability of belonging to the at-risk cluster rather than a hard label.

4. Use of Tools: Python with pandas for aggregating monthly usage data, scikit-learn's KMeans (and GaussianMixture as an optional refinement) for clustering, and a dashboard (e.g., matplotlib/Power BI) plotting usage vs. recharge to let the retention team visually monitor at-risk clusters over time.

5. Analysis & Business Impact: U2 and U4 would be flagged as churn-prone and proactively targeted with retention actions — discounted recharge plans, bonus data, or personalized customer-care outreach — while U1, U3, U5 continue receiving loyalty/upsell offers instead. Tracking cluster membership month-over-month also reveals users migrating from the engaged cluster toward the at-risk cluster, giving early warning before they actually churn.

Q4. E-commerce Behavior — Improving Recommendation Systems

Dataset:

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question: Suggest a clustering technique to group users and explain how it improves recommendation systems.

Answer:

1. Understanding the Problem: An e-commerce platform wants to recommend products more effectively by understanding how actively different users browse and convert. Grouping users with similar engagement funnels (views → clicks → purchases) allows the recommendation engine to tailor strategy per group instead of treating all users identically.

2. Application of Engineering Concepts: U1, U3 and U5 show high views (50–60), high clicks (20–25) and multiple purchases (5–6) — highly engaged, high-converting users. U2 and U4 show low views (20–30), low clicks (8–10) and minimal purchases (1–2) — browsing but rarely converting. This is a natural two-cluster engagement pattern.

3. Solution Design: Normalize the three behavioural features and apply K-means clustering with $k = 2$ to separate 'Highly Engaged Buyers' from 'Passive Browsers.' Because the three features are highly correlated (more views generally lead to more clicks and purchases), Principal Component Analysis (PCA) can optionally be applied first to reduce dimensionality and denoise the clustering input before running K-means.

4. Use of Tools: Python with scikit-learn (KMeans, optionally PCA), pandas for building the view-click-purchase feature table per user, and visualization via a scatter plot of clicks vs. purchases coloured by cluster to validate the grouping.

5. Analysis & Business Impact: For 'Highly Engaged Buyers' (U1, U3, U5), the recommendation system can surface complementary/premium products (cross-selling) and personalized bundles, since these users are primed to purchase. For 'Passive Browsers' (U2, U4), the system can instead show best-seller or highly-rated items, limited-time discounts, or simplified recommendations to nudge them from browsing to converting. This cluster-aware recommendation strategy improves click-through and conversion rates compared to a single generic recommendation model applied to all users.

Q5. Traffic Data — Identifying Congestion Zones

Dataset:

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question: Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management.

Answer:

1. Understanding the Problem: Traffic authorities need to identify which road locations are genuinely congested — characterized by high vehicle count, low average speed, and long delays — versus

locations flowing normally, so that traffic signals, diversions, and infrastructure investment can be prioritized correctly.

2. Application of Engineering Concepts: L2 and L4 show high vehicle counts (500–600), low average speed (10–15 km/h) and long delays (25–30 min) — clear congestion hotspots. L1, L3 and L5 show lower vehicle counts (200–220), higher speed (28–32 km/h) and short delays (9–12 min) — free-flowing locations. The three features move together in a consistent way, making this a well-defined two-cluster problem.

3. Solution Design: Standardize vehicle count, average speed, and delay, then apply K-means clustering with $k = 2$ to separate 'Congested Zones' from 'Free-Flowing Zones.' Since congestion severity can vary continuously rather than being strictly binary across a larger real-world dataset with many more locations, the Elbow Method or Silhouette Score should be used to validate whether more than two clusters (e.g., mild/moderate/severe congestion) better fit the full city-wide dataset; for a larger, non-uniformly distributed set of locations across a city, density-based clustering (DBSCAN) is also worth considering to automatically detect irregularly-shaped congestion corridors along road networks rather than assuming compact, spherical zones.

4. Use of Tools: Python with pandas for aggregating sensor/GPS data per location, scikit-learn's KMeans (and DBSCAN as an alternative), and a geographic heat map (e.g., using folium or GIS software) overlaying the cluster labels on a city map for traffic control-room visualization.

5. Analysis & Business Impact: L2 and L4 would be flagged as priority congestion zones for interventions such as adaptive traffic-signal timing, rerouting suggestions pushed to navigation apps, or long-term infrastructure upgrades (flyovers, additional lanes). L1, L3, L5 require no immediate intervention and can be monitored at lower priority. This clustering-driven approach lets traffic management authorities allocate limited resources (traffic police, signal upgrades, budget) to the zones where they will have the greatest impact on reducing city-wide congestion.